

Grounding or Lexical Overlap?

A Matched-Pair Study of Internals-Based Faithfulness Probes

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Abstract

We provide a confound-controlled methodology and a released “wrong-question” diagnostic for internals-based, reference-free hallucination detection, which treats *context-faithfulness* (is an answer entailed by the context?) as a linear residual-stream direction. In the standard extractive setup, faithfulness is almost perfectly correlated with *lexical overlap* ($r=0.99$), so an apparent “grounding probe” may be a string-matching detector. Our matched SUPPORT×OVERLAP construction *partially* decouples the two (halving the confound to $r=0.50$), and the diagnostic fools an off-the-shelf NLI model. Across four instruction-tuned 7–9B families: (i) within the overlap-controlled stratum a support contrast is linearly decodable, but a *no-context ablation* shows it reflects question–answer compatibility rather than context use, and is inseparable from factoid-correctness by construction; (ii) it aligns across families up to a linear map, yet answer-length, overlap, and factuality directions align equally well, a generic matched-stimulus property, not grounding-specific; (iii) on the three families we could extract, the synthetic direction does not transfer to real hallucinations, scoring below NLI, confidence, and lexical overlap on RAGTruth, where only an in-domain probe trained on human labels recovers a usable signal.

1 Introduction

A grounded language model should answer from the *provided context*, not from its parametric memory or from surface cues. Reference-free, internals-based methods operationalize this by reading a linear direction from the residual stream: much prior work decodes a “truth” or “lying” direction (Azaria and Mitchell, 2023; Marks and Tegmark, 2024; Bürger et al., 2024), and more recent work extends this style of analysis to a *contextual* faithfulness direction (O’Neill et al., 2025; Gao et al., 2026),

an attractive recipe because it needs no external judge and costs roughly one forward pass. We interrogate this recipe *family* rather than any single system: we probe a representative mass-mean direction rather than reproducing a specific observer-model detector, so our negatives bound *this* synthetic axis (§5.4), not stronger internals detectors such as ReDeEP (Sun et al., 2025), which we do not test. Two distinct notions are routinely conflated under this banner: *parametric-factuality* (is a claim true in the model’s memory?) and *context-faithfulness* (is a claim entailed by the given context?). Recent work treats both as activation objects and asks how they relate (Wang et al., 2025), but samples the two notions from *different* datasets, so any measured geometry is confounded with topic and lexicon.

We identify a sharper confound that undermines the faithfulness side specifically. In the standard extractive QA setup used to build such probes (e.g. entity-swap contexts (Longpre et al., 2021)), an answer is context-faithful *iff* the answer string occurs in the context. Faithfulness is therefore almost perfectly correlated with *lexical overlap* ($r=0.99$ in our data), and a probe that appears to read “grounding” may simply detect string copying: because faithfulness and overlap nearly coincide, a near-ceiling probe *could* be reading either, and telling them apart is what our matched construction (§3) is built to do.

A matched control. We introduce a SUPPORT×OVERLAP construction (§3) that crosses semantic support with lexical presence in a 2×2 of which we realize three cells, all model-free: the matched supported-present cell, a *wrong-question trap* where the answer is present but answers a different question (supported=no, overlap=yes), and the usual unsupported-absent cell. A paraphrase cell would close the design but has zero yield in this release (§3). This drops

$\text{corr}(\text{SUPPORT}, \text{OVERLAP})$ from 0.99 to 0.50, and, tellingly, an off-the-shelf NLI model is fooled by the clean wrong-question trap $\sim 25\%$ of the time, confirming that lexical overlap misleads even “gold” entailment checks.

Contributions. With this control we run a careful study across four model families (Llama-3.1-8B, Qwen2.5-7B-Instruct, Mistral-7B-v0.3, Gemma-2-9B). The result is a confound-controlled methodology and a series of controls that, one by one, attribute the apparent “grounding” signal to factors other than context use.

1. **An overlap-controlled internal signal that reflects question–answer compatibility, not context-grounding (§5.1).** Within the answer-present stratum (lexical overlap held constant), a support contrast is decodable on all four families (collision-free raw AUROC near-ceiling, ≈ 0.99 ; 0.76–0.88 after purging model confidence). But a *no-context ablation* shows it is mostly question–answer compatibility: removing the context barely changes it (raw 0.95–0.98, confidence-purged does not collapse), so the probe separates the matched answer from the wrong-question answer *without reading the context*. Support also coincides with our *synthetic* factuality label here by construction, so factoid-correctness is not excluded either.
2. **Cross-family alignment up to a linear map (§5.3).** An orthogonal-Procrustes map carries the support direction between families so the transported direction is nearly collinear with the target family’s *native* direction (cosine ≈ 0.98 , vs. ≈ 0.01 for a random direction through the same map), whereas anchor-projection transfer (Kim and Han, 2026) reaches only 0.76 AUROC here. The transferred-probe AUROC (0.901) equals the within-family bound but is ceiling-saturated, and signed map-aware nulls sit at chance (0.51/0.54); we therefore rest the claim on the cosine. Factuality, answer-length and overlap directions transport identically while a confidence direction does not, indicating a shared stimulus-derived contrast rather than a grounding-specific axis; we do not generalize beyond these four 7–9B models.
3. **The confidence asymmetry runs the wrong way (§5.2).** Contrary to a natural hypothesis, confidence absorbs the *support* axis more than the factuality axis at every one of 71 layer-tests

(mean -0.142), so confidence-purging does not separate the two notions.

4. **Synthetic axes do not transfer to real hallucinations (§5.4).** Applied to RAGTruth (Niu et al., 2024), the controlled-stimulus support direction scores *below* NLI, confidence, and lexical overlap among single-pass API-free baselines; within a task no *label-free* internals- or NLI-based signal, including a sentence-level SummaC-style NLI baseline, exceeds ≈ 0.6 macro AUROC, and even lexical overlap, the strongest baseline, reaches only macro 0.647. An in-domain probe trained on RAGTruth human labels recovers a usable signal (0.76–0.81 on the three RAGTruth families; exceeds a conservative single-hypothesis NLI baseline but ties lexical overlap on Qwen), so a real support signal exists in the internals but is a *different direction* from the synthetic axis and needs in-domain annotation.

The takeaway for the “faithful and efficient” agenda is twofold: internals-based support contrasts identified on controlled data are real and remarkably portable across model families (though not grounding-specific), but are not, by themselves, deployable hallucination detectors: on real data, lexical overlap still dominates.

2 Related Work

Geometry of truth and faithfulness. Truthfulness is decodable from activations (Azaria and Mitchell, 2023), largely linearly (Marks and Tegmark, 2024), and partly shared across models (Bürger et al., 2024); how far such a direction transfers is contested. Azizian et al. (2025) find the geometries of truth near-orthogonal across tasks, whereas Bao et al. (2025) report generalization from atomic statements to question answering in the more capable models they test. Wang et al. (2025) treat factuality and faithfulness hallucinations as activation subspaces and argue they overlap; crucially they draw the two notions from different corpora, leaving the geometry confounded with surface form. Adarsh et al. (2026) study how a single truth vector rotates when context is added. We instead hold the two notions on *matched* stimuli and control lexical overlap explicitly; we note, however, that our factuality label is a synthetic gold-answer relabeling rather than a parametric memory-truth probe, so we engage this line methodologically without claiming to adjudicate whether factu-

ality and faithfulness share a direction.

Knowledge conflict and source attribution.

Probes can detect knowledge conflict and predict which source a model follows (Zhao et al., 2024; Sun et al., 2025; Brink et al., 2026), decoding-time interventions can steer a model toward the context (Shi et al., 2024), and benchmarks such as FaithEval (Ming et al., 2025) score context-versus-parametric faithfulness behaviorally. These are behavioral/source signals; we ask whether a *content* grounding axis exists that is not reducible to lexical overlap.

Cross-model transfer. Linear concept directions can be aligned across models via orthogonal Procrustes (Schönemann, 1966; Puri et al., 2025) or anchor projection (Kim and Han, 2026). We apply both to a support direction; Procrustes renders it nearly collinear with the target’s native direction (cosine ≈ 0.98), whereas anchor projection reaches only 0.76 transfer AUROC (§5.3).

Reference-free hallucination detection.

Reference-free detectors fall into three families: *external* entailment/alignment scorers (Zha et al., 2023); *internals*-based probes (single residual directions (O’Neill et al., 2025) and the truth/faithfulness directions above); and *confidence*/correctness signals (Cho et al., 2026; Moreno Cencerrado et al., 2025), with faithfulness-aware UQ (Fadeeva et al., 2026) and self-aware RAG abstention (Zhu et al., 2026) combining several. We use one baseline from each family on RAGTruth (external NLI, model confidence, and lexical overlap), and find our controlled-stimulus axis does not beat them.

3 The SUPPORT $\times$ OVERLAP Construction

We build matched statements from NQ-Swap (Longpre et al., 2021), which provides, per question q , a gold answer a_{true} supported by an original context c_{org} and a substituted entity a_{swap} supported by an entity-swapped context c_{sub} . We define three bits per (context, answer): SUPPORT S (does the given context entail the asserted answer for the asked question?), OVERLAP O (does the answer string occur in the context?), and FACTUALITY F (is the asserted answer the correct answer to the asked question, irrespective of the context?). S and F coincide except in the answer-absent cell D, where the answer is still factually right but the substituted context no longer supports it. The 2×2

design has four cells, of which we realize three (cell B is unrealized in this release; see below):

cell	(question, context, answer)	S	O	F
A	$(q, c_{\text{org}}, a_{\text{true}})$	1	1	1
B	$(q, \text{paraphrase}(c_{\text{org}}), a_{\text{true}})$	1	0	1
C	$(q', c_{\text{org}}, a_{\text{true}})$	0	1	0
D	$(q, c_{\text{sub}}, a_{\text{true}})$	0	0	1

Cell C is a *wrong-question trap*: a_{true} is present in c_{org} but is offered as the answer to a different question q' (the next item’s question). A realized triple:

A	$(q, c_{\text{org}}, a_{\text{true}})$	$S=1, O=1, F=1$
	q : who played maudie in the movie el dorado	
	c : Charlene Holt as Maudie; a : Charlene Holt	
C	$(q', c_{\text{org}}, a_{\text{true}})$	$S=0, O=1, F=0$
	q' : who won the american league pennant in 2017	
D	$(q, c_{\text{sub}}, a_{\text{true}})$	$S=0, O=0, F=1$
	c_{sub} : Gwendoline Christie as Maudie	

The donor q' is unrelated in topic, so the A/C contrast is in principle solvable from question–answer compatibility alone, without consulting the context at all: a possibility §5.1 tests directly and confirms. Cell B would paraphrase c_{org} so the answer is entailed but not copied (open instruct model, keeping an item only if token-overlap ≤ 0.34 and an NLI model still entails it); the released build supplies no paraphraser, so cell B has *zero* yield and all reported analyses use the model-free cells A/C/D with $\text{corr}(S, O)=0.50$ (0.56 after removing the donor-collision items below; an orthogonalized $\text{corr}\approx 0$ design would require cell B). One caveat in cell C: NQ-Swap dev has only 418 distinct questions over 1,200 items, so for 278 items (23.2%) the donor q' is string-identical to q and a_{true} *does* answer it; we flag these *donor-collision* items and report cell-C results with them removed (the full construction will dedup $q'\neq q$). On *clean* cell C an off-the-shelf NLI checkpoint (Laurer et al., 2024; He et al., 2023) still rates 25.5% of unsupported-but-present items as “supported” (mean entailment 0.31, vs. 0.74 for cell A and 0.18 for cell D); the raw 37.6% rate is inflated by the collisions, which NLI entails 77.7% of the time (correctly). Lexical overlap thus misleads “gold” entailment even after the label error is purged (Fig. 1).

4 Probing, Separability, and Transfer

Directions. For a (model, layer) we extract last-token residual states X over the asserted answer span (one forward pass, bf16, left-padded). A SUPPORT direction d_S and FACTUALITY direction d_F are mass-mean (difference-of-means) estimates

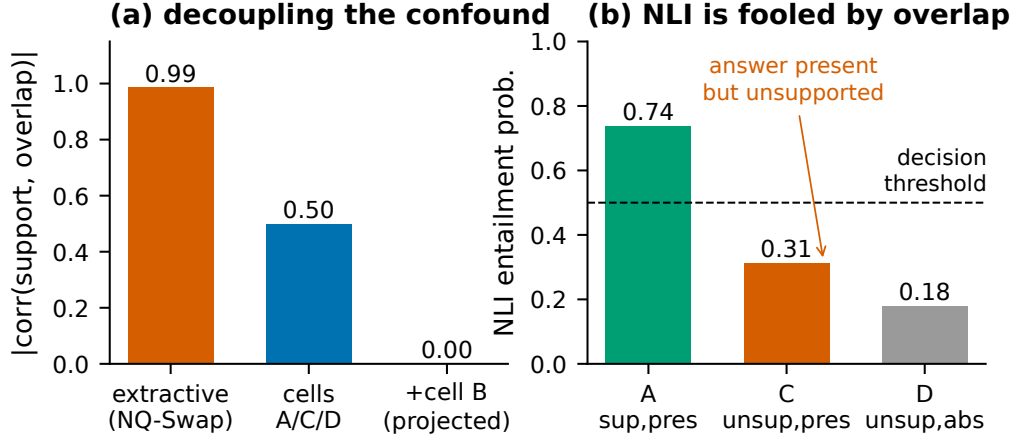


Figure 1: **Partially decoupling support from lexical overlap.** (a) The matched construction drops $|\text{corr}(\text{SUPPORT}, \text{OVERLAP})|$ from 0.99 (extractive NQ-Swap) to 0.50 (model-free cells A/C/D), and would reach ≈ 0 with the paraphrase cell B (projected; cell B has zero yield in this release). (b) Mean NLI entailment by *clean* cell: on the wrong-question trap (C), where the answer is present but unsupported, NLI is still pulled toward “supported.”

(Marks and Tegmark, 2024); signed projections Xd are the probe scores. Every reported number uses one protocol. Folds are grouped by *source*, not by record: by the normalized (question, context, answer) triple of the underlying NQ-Swap QA for CtrlPairs (1,200 records, 555 triples) and by source passage for RAGTruth (2,000 responses, 450 passages), since ITEM_ID indexes one swap record or one response and does not isolate sources. Residualization is fitted *within* each training fold. AUROC is *signed*: polarity is fixed on the training fold and the test projection scored as-is, with no $\max(\text{AUC}, 1 - \text{AUC})$ flip anywhere, so a non-predictive direction averages to 0.5 instead of being floored above it. Folds come from a seeded group splitter; pooled estimates are $\text{mean} \pm \text{sd}$ over ten splits, with source-cluster bootstrap CIs on the first.

Overlap control. To test grounding *beyond* overlap we (i) restrict to the answer-present stratum $O=1$ (cells A vs. C), where lexical presence is constant, and (ii) residualize X on surface covariates (overlap, length) before re-probing.

Null-calibrated separability. We treat each notion as an undirected axis and measure the acute angle $\angle(d_S, d_F)$. Significance is assessed against a within-notion null built by splitting one notion’s data in half, re-estimating the same axis on each half and measuring the half-vs-half angle; this is a resampling null, not a label permutation. With $n_{\text{null}}=2,000$ draws per notion we report the reso-

lution bound $p < 5 \times 10^{-4}$ rather than $p=0$ when no null draw exceeds the observed angle. It is run per family at the a-priori mid layer on all three cells; it cannot be posed *within* the overlap-controlled stratum, where S and F coincide exactly.

Confidence purge. We regress each hidden dimension on three confidence covariates (mean log-prob, mean max-softmax, first max-softmax) and re-probe the residual. For the asymmetry test (§5.2) we quantify how much of an axis *is* confidence as absorbed, the adjusted R^2 of that axis’s projection regressed on the same three covariates; unlike a purge-then-re-probe AUROC delta this is not confounded by the denoising side effect of residualizing.

Cross-family transfer. Families differ in hidden size, so we fit a semi-orthogonal Procrustes map between two models’ residual spaces on a shared calibration split, transport a source-trained probe, and measure target AUROC. The calibration/evaluation split is a seeded half-split over source-QA groups (items in the calibration set are disjoint from the evaluation set), so the map is never fit and tested on near-duplicate rows of the same item; transported-probe polarity is fixed on the target calibration fold. We compare to the anchor-projection ACS baseline (Kim and Han, 2026), a within-family upper bound, and a random-direction floor (also scored without the polarity flip, so it sits at chance rather than being inflated above 0.5).

5 Experiments

Setup. Four instruction-tuned families: Llama-3.1-8B (Llama Team, 2024), Qwen2.5-7B (Qwen Team, 2024), Mistral-7B-v0.3 (Jiang et al., 2023), Gemma-2-9B (Gemma Team, 2024). CtrlPairs: 1,200 NQ-Swap items $\rightarrow$ 3,593 matched statements (cells A/C/D). RAGTruth: 2,000 test responses with response-level hallucination labels (Niu et al., 2024). Baselines: NLI (the DeBERTa-v3-base-mnli-fever-anli checkpoint (Laurer et al., 2024), DeBERTaV3 (He et al., 2023) fine-tuned on MNLI, FEVER and ANLI (Williams et al., 2018; Thorne et al., 2018; Nie et al., 2020)) in two variants, (i) *whole-response*: a single (context, declarativized-answer) entailment query; and (ii) *sentence-level*, a SummaC-style decomposition (Laban et al., 2022) that entailment-checks each response sentence against every context sentence, takes the max over context, and aggregates over response sentences by mean (SummaC-ZS) or min (for tractability we cap context at 60 sentences and pairs at 256 tokens, which can only lower SummaC and is thus conservative), plus max-softmax confidence and lexical overlap. Everything is API-free.

5.1 The within-stratum signal is question–answer compatibility, not grounding

Leakage control. Grouping on ITEM_ID is not sufficient, because it isolates swap records, not source QAs: 941 of the 1,200 cell-A rows have an identical twin (after case/whitespace normalization) under a different ITEM_ID (mid-layer cosine 0.9999), and 903 of them landed in a different fold. We group by the source QA triple instead (555 groups), for the C1 cross-validation, the Procrustes calibration split (§5.3) and the cross-strata control, which is additionally evaluated on *held-out* groups. Regrouping moves every C1 estimate by at most 0.007 (App. A): the leak is real but inert. To avoid selecting the layer on the very number we report, we fix the analysis layer *a priori* at relative depth 0.5 (the nearest cached layer to $0.5 L_{\text{tot}}$: $L=14/16/16/21$ for Qwen/Mistral/Llama/Gemma), matching the relative-depth convention already used for C3/C4 and for Fig. 2; App. A reports the mean $\pm$ sd across the mid-layer band (0.4–0.6 depth).

Table 1 reports support decodability within the answer-present stratum $O=1$ (cells A vs. C), where

lexical overlap is held constant, on the *collision-free* cells: we drop the 23.2% of cell-C items whose donor question is string-identical to the original (§3), which would otherwise mislabel genuinely-supported items as unsupported. Four observations bound the interpretation. First, the raw within-stratum AUROC is *near-ceiling* (0.99; 0.988–0.998 across families): once mislabels are removed, *A* and *C* are almost perfectly separable. We read this less as strong grounding than as evidence that the contrast is *easy*: within $O=1$ it varies *only the question* (context and answer are fixed), so a probe can in principle separate *A* from *C* by question–answer appropriateness without using the context. Second, the operative confound is *model confidence* (overlap and answer length are uninformative here, AUROC 0.500): answer log-prob and NLI separate *A* from *C* on their own, so we residualize the hidden states on the three confidence scalars; the support contrast then falls to 0.76–0.88, which we take as the primary confound-controlled estimate. Third, and decisively, a *no-context ablation* settles what the signal is (Table 2): re-extracting the *A/C* activations with the context *removed* (question+answer only), or with it replaced by an unrelated passage, leaves $\text{supp}_{O=1}$ almost unchanged, and the confidence-purged estimate does not collapse either, and it *rises* for three of four families, and rises further under a substituted context. The within- $O=1$ contrast is therefore mostly *question–answer compatibility*: the probe separates *A* (whose question the answer fits) from *C* (whose donor question it does not) *without reading the context at all*. A cross-strata control (train *A* vs. *D*, which fixes the question but also varies overlap since *D* is answer-absent; evaluate *A* vs. *C*) does transfer at 0.66–0.88. Here *A* and *D* share the question and the answer string and differ only in the context, so an *A*-vs-*D* direction cannot be a question–answer compatibility axis by construction (the exact complement of the ablation above), and *some* context-sensitive component therefore exists, but it is an upper bound and not separable from the dominant compatibility signal here. Finally, support and our *synthetic* factuality label are *identical* within $O=1$ by construction (they diverge only in the answer-absent cell D). We therefore make no context-grounding claim within this stratum.

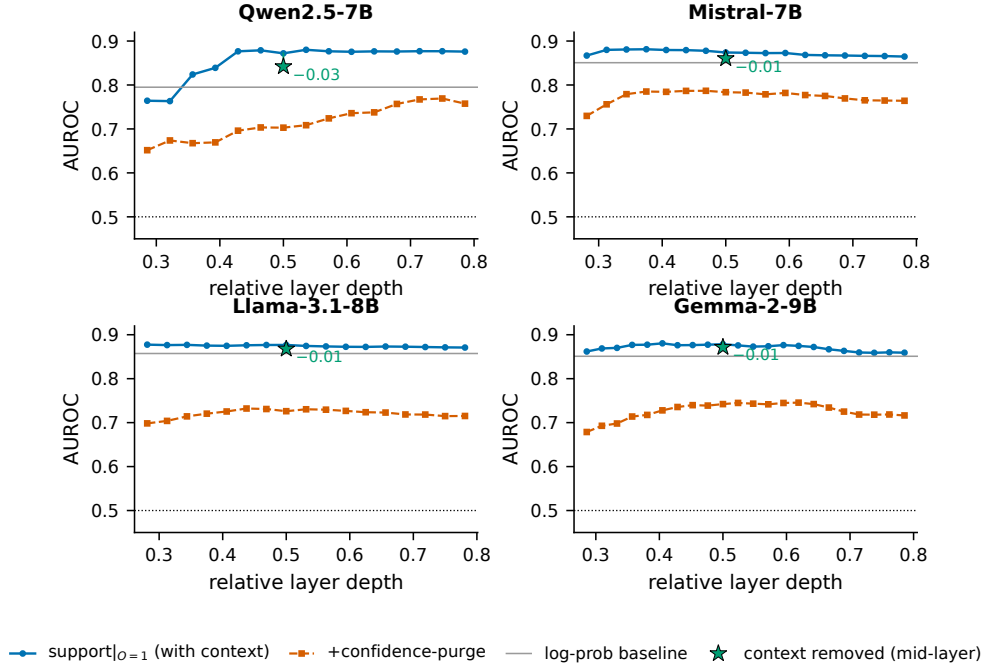


Figure 2: **The within-stratum contrast peaks mid-layer, but it is not grounding.** Cross-validated AUROC vs. relative layer depth for the support contrast within the answer-present stratum ($O=1$, where overlap is uninformative): raw, and after purging model confidence; the gray line is a log-probability baseline. The contrast is high and stable through mid layers on all four families. Curves use the same source-grouped, signed protocol as Table 1 but *retain* the donor-collision mislabels, hence their lower level (≈ 0.87 raw); removing those items is what lifts the contrast to near-ceiling. The green star marks the same probe at the a-priori layer with the context *removed*: the raw AUROC barely moves (drop annotated), and shuffled context behaves the same. On the collision-free sample of Table 1 (not the curves plotted here) the purged value even *rises* for Qwen ($0.760 \rightarrow 0.838$) and Llama, is flat for Gemma, and falls materially only for Mistral ($0.878 \rightarrow 0.805$). The within- $O=1$ contrast is therefore mostly question–answer compatibility rather than context use.

5.2 The confidence asymmetry runs the wrong way

On the full three-cell sample the two axes are not collinear: at the a-priori layer $\angle(d_S, d_F)$ is $78^\circ/54^\circ/67^\circ/65^\circ$ for Qwen/Mistral/Llama/Gemma, each beyond every draw of its within-notion null ($p < 5 \times 10^{-4}$). This says little, though: those stimuli have $\text{corr}(S, F) = 0.50$ by construction, and inside the overlap-controlled stratum the labels coincide exactly, so the angle cannot establish an overlap-independent axis.

We also hypothesized that confidence absorbs factuality more than faithfulness, which would separate them. It does not, and the effect runs the other way at every layer we measured. Over all four families (71 layer-tests), $\text{absorbed}(F) - \text{absorbed}(S)$ averages -0.142 (sd 0.062): *all* 71 are negative and 70 are one-sided significant in the direction *opposite* to the hypothesis (per-family means -0.104 , -0.122 , -0.131 , -0.193), so confidence absorbs the support axis *more* and we do not rely on

confidence-purging to distinguish the two notions.

5.3 Matched-stimulus contrasts align across families up to a linear map

We ask whether the support direction occupies the *same* geometry across families up to a linear map. The direct evidence is geometric: the Procrustes-mapped source direction is nearly collinear with the target family’s *native* direction (transported-to-native cosine 0.976 , range 0.95 – 0.99 over the 12 ordered pairs; Table 3); anchor projection reaches only 0.76 transfer AUROC against Procrustes’ 0.901 (Fig. 3), and the two are not comparable in cosine because ACS has no target-space direction. The cosine is far above its null: a random or label-shuffled source direction transported through the *same* fitted map is essentially orthogonal to the native direction (cosine 0.01 and 0.13 respectively), so 0.976 is not an artifact of the shared-stimulus geometry. It is also not an artifact of fitting the map and the directions on the same items: with

model	L	$\text{supp}_{O=1}$	+conf	cross
Qwen2.5	14	0.988	0.760	0.661
Mistral	16	0.996	0.878	0.870
Llama	16	0.997	0.789	0.876
Gemma	21	0.998	0.811	0.827

Table 1: **C1 (confound-controlled, collision-free)**. Layer fixed a priori at relative depth 0.5; folds grouped by *source QA* (555 groups over 1,200 records); polarity resolved on the train fold and scored signed; donor-collision cell-C mislabels removed (§3; $n_A=1,200$, $n_C=922$). Each entry is the mean over ten seeded splits ($\text{sd} \leq 0.003$). $\text{supp}_{O=1}$ is support AUROC within the answer-present stratum, near-ceiling once mislabels are removed, because the contrast varies only the question (source-cluster 95% CIs [0.983, 0.992], [0.993, 0.998], [0.995, 0.999], [0.997, 0.999]). +conf: hidden states residualized on the three confidence scalars, fitted within each fold; our primary estimate. cross: train A vs. D , evaluate A vs. C on *held-out* groups, *raw* rather than confidence-purged, hence an upper bound and not on the same footing as the +conf column. Within $O=1$ support and the synthetic factuality label coincide, so we make no distinctness claim.

model	L	with context		no context		shuffled	
		raw	+conf	raw	+conf	raw	+conf
Qwen2.5	14	0.988	0.760	0.947	0.838	0.938	0.888
Mistral	16	0.996	0.878	0.975	0.805	0.976	0.809
Llama	16	0.997	0.789	0.980	0.814	0.982	0.839
Gemma	21	0.998	0.811	0.984	0.812	0.983	0.841

Table 2: **The within- $O=1$ contrast does not need the context.** $\text{supp}_{O=1}$ re-extracted with the context removed (question+answer only) and with it replaced by an unrelated passage; same a-priori layers and same collision-free sample as Table 1 ($n_A=1,200$, $n_C=922$), same source-grouped signed protocol. Removing the context costs at most 0.05 raw, and the confidence-purged contrast is *higher* without the context for three of four families, and highest of all when the context is replaced by an unrelated passage (Qwen 0.760→0.888). Only Mistral falls (0.878→0.805).

the map fit on a calibration split and the target’s *native* direction re-estimated on a disjoint, source-grouped held-out split, the transported-to-native cosine is 0.962 ± 0.015 (mean over 12 pairs $\times 10$ splits; 0.958 with both source and target directions held out from the map), close to the in-sample value (Table 3); the ~ 0.014 drop is about the native direction’s own cross-split self-consistency loss (0.986), so the Procrustes map adds no measurable inflation (factuality behaves identically, 0.990→0.975). We deliberately do *not* headline the transferred-probe AUROC (0.901): it equals the within-family bound

axis	cosine	AUROC	within	rand-map	rand-lab
SUPPORT	0.976	0.901	0.900	0.509	0.543
FACTUALITY	0.990	0.914	0.910	—	—
<i>nuisance directions:</i>					
length	0.977	—	—	—	—
overlap	0.991	—	—	—	—
confidence	0.509	—	—	—	—

Table 3: **C3 (cosine-based, leakage-controlled)**. Mean over the 12 cross-family ordered pairs and ten seeded source-grouped splits. *cosine*: transported-to-native cosine (the alignment evidence; non-saturating); shown in-sample, it stays at 0.962 when the native direction is re-estimated on a source-grouped held-out split (§5.3), so it is not a map-fitting artifact. A random / label-shuffled source direction through the *same* map has cosine only 0.01 / 0.13 with the native direction. *AUROC/within*: transferred-probe vs. within-family AUROC: equal, hence ceiling-saturated. *rand-map/rand-lab*: random / shuffled-label directions through the map, signed AUROC, at chance. Support, factuality, answer-length and overlap directions all align at ≈ 0.98 , while a model-internal confidence direction does not (0.51): alignment is a generic property of contrasts derived from the shared stimuli, not a grounding-specific axis.

(0.900) and is therefore ceiling-saturated; scored with polarity-honest (signed) AUROC, random and shuffled-label directions through the map sit near chance (0.51/0.54). **This alignment is not specific to grounding.** Transporting the *factuality*, *answer-length*, and *lexical-overlap* directions through the same maps gives cosines of 0.99/0.98/0.99, as high as support, whereas a purely model-internal *confidence* direction transports at only 0.51 (and unstably, 0.08–0.87 across pairs). So what aligns is any contrast *derived from the shared input stimuli*, a shared low-rank stimulus geometry, not a grounding-specific axis; that confidence does *not* align shows the alignment is non-trivial but also that it carries no grounding-specific content. We therefore conclude only that *matched-stimulus contrasts, support among them, are linearly alignable across these four 7–9B families*, and avoid any claim of a grounding-specific or near-lossless transfer.

5.4 Synthetic axes do not transfer to real hallucinations

We learn d_S on CtrlPairs and apply it to the fixed RAGTruth responses (the NLI and overlap baselines are response-intrinsic, hence identical across models), predicting the human faithfulness label (Table 4). Here we keep the *support-polarity* sign

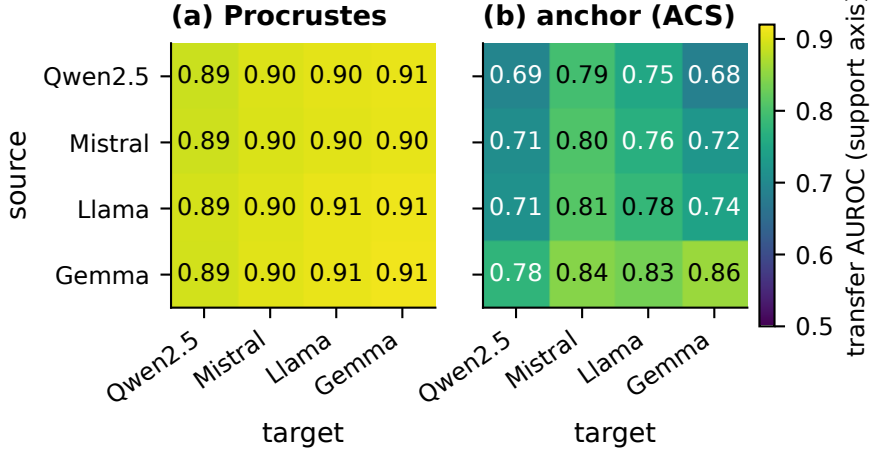


Figure 3: **Matched-stimulus contrasts align across families up to a linear map.** Source→target transfer AUROC of the support direction; every cell, diagonal included, uses that panel’s own estimator, source-grouped. (a) Orthogonal Procrustes is uniformly high on- and off-diagonal (0.89–0.91); we read this as alignment (cosine 0.98, Table 3), not lossless transfer, since the AUROC is ceiling-saturated. (b) Anchor projection (ACS), scored the same way, is uniformly lower (0.68–0.86) and its own diagonal (0.69–0.86) stays well below Procrustes’ 0.89–0.91, so its gap to Procrustes is not a transfer effect.

fixed from CtrlPairs and score without the test-fold flip, because flipping per-test inflates a near-chance axis; CtrlPairs and RAGTruth share no items, so there is no leakage in this transfer. Scored honestly, the label-free synthetic axis reaches only 0.54–0.66 AUROC (and drops toward chance for the weakest family), *below* NLI (0.673), model confidence (0.62–0.72), and plain lexical overlap (0.753), the strongest label-free baseline on RAGTruth (Fig. 4). Paired source-passage cluster-bootstraps confirm the ordering against overlap ($\Delta = -0.097 / -0.163 / -0.212$) and confidence ($p < 0.001$, $p = 0.028$, $p < 0.001$), though against whole-response NLI only Llama and Gemma separate ($p < 0.001$); Qwen ties ($\Delta = -0.017$, CI $[-0.051, +0.017]$). A more sophisticated sentence-level NLI baseline (SummaC-style, max entailment over context sentences) does no better: pooled 0.614/0.664 and within-task macro 0.596/0.579 (mean/min), essentially tied with the whole-response NLI (0.582) and still well below overlap (Table 6); many RAGTruth hallucinations are abstractive errors that survive sentence-level entailment. An *in-domain* probe (5-fold grouped-CV L_2 -logistic on the same mid-layer states, target=human faithfulness label) does recover a usable signal: AUROC 0.76/0.81/0.80 for Qwen/Llama/Gemma (passage-cluster 95% CIs in Table 4). It exceeds our NLI baseline (0.673) on all three families (a conservative whole-response check, §5); against the strong overlap

baseline (0.753) it wins on Llama ($p = 0.002$) and Gemma ($p < 0.001$, paired source-passage cluster bootstraps) but only ties on Qwen ($+0.01$, $p = 0.27$). Residualizing lexical overlap and answer length out of the features first, within each training fold, leaves 0.68–0.73, so some of this signal is overlap but much of it is not. A real support signal therefore exists in the internals, but it is a *different direction* from the synthetic axis and is recoverable only with in-domain human annotation. That comparison changes the estimator (mass-mean to L_2 -logistic) and the training domain at once, so we separate them (Table 5): holding the estimator at mass-mean, the *same* estimator that transfers at 0.596 from CtrlPairs reaches 0.781 fitted in-domain, whereas swapping mass-mean for L_2 -logistic moves it by at most 0.074 and in both directions. The two effects do not overlap: every domain effect in the table (0.131–0.306) exceeds every estimator effect (0.016–0.074). What separates the two is the training domain, not the estimator.

6 Conclusion

Our results separate two questions usually merged. *Does* a non-overlap signal exist in the internals? Within the overlap-controlled stratum a contrast is decodable (0.76–0.88 purged) and aligns across four families up to a linear map, but a no-context ablation shows it is mostly *question–answer compatibility*, not context use, and the alignment is

model	label-free		labeled in-dom [†]	NLI	overlap
	synth- d_S	conf			
Qwen2.5	0.657	0.723	0.765	0.673	0.753
Llama	0.591	0.623	0.814	0.673	0.753
Gemma	0.541	0.686	0.797	0.673	0.753

Table 4: **C4 (leakage-controlled)**. RAGTruth hallucination detection (AUROC; three families, Mistral’s RAGTruth extraction OOM’d). The label-free synthetic support axis (synth- d_S , support polarity fixed from CtrlPairs, a-priori mid layer, scored without the test-fold flip) is below NLI, confidence, and lexical overlap, and for the weakest family (Gemma, 0.541) is barely above chance. [†]uses RAGTruth human labels (not label-free); source-passage cluster-bootstrap 95% CIs [0.737, 0.786], [0.780, 0.824], [0.776, 0.820].

model	CtrlPairs-trained		RAGTruth-trained	
	mass-mean	logistic	mass-mean	logistic
Qwen2.5	0.657	0.623	0.788	0.765
Llama	0.591	0.517	0.774	0.813
Gemma	0.541	0.492	0.781	0.798
mean	0.596	0.544	0.781	0.792

Table 5: **Estimator $\times$ training domain (C4)**. RAGTruth AUROC at the same a-priori layers, scored signed. Column 1 is the synth- d_S column of Table 4; the other three hold one factor fixed in turn. Changing the training domain is worth ≈ 0.19 at a fixed estimator (mass-mean 0.596 $\rightarrow$ 0.781); changing it is worth at most 0.074 and moves in both directions: CtrlPairs-trained logistic is *worse*, and below chance on Gemma. The ranges are disjoint: domain effects span 0.131–0.306, estimator effects 0.016–0.074. Both RAGTruth columns are 5-fold passage-grouped CV over 20 seeds ($sd \leq 0.006$); the logistic one is the same quantity as Table 4’s in-domain column and agrees with it to 0.001.

a generic matched-stimulus property (length and overlap directions align equally well). An apparent “grounding” signal thus survives an overlap control yet is still not grounding. *Is that direction a deployable, label-free hallucination detector?* No, it does not transfer to RAGTruth, where lexical overlap remains the strongest single-pass baseline (and even that edge is largely a between-task base-rate artifact: Table 6). The gap between a clean controlled signal and real-world utility is the main lesson, and it cautions against reading deployment claims from synthetic-stimulus probes; whether any label-free signal can beat lexical overlap on real hallucinations is the natural next question.

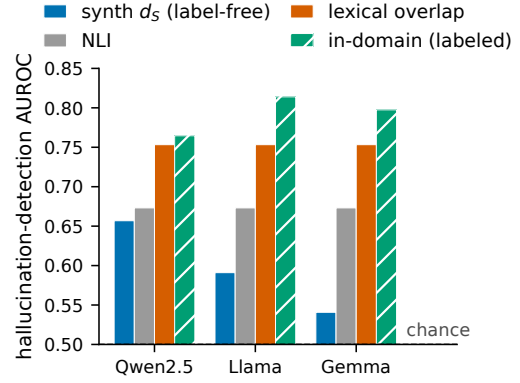


Figure 4: **The synthetic axis does not transfer to real hallucinations.** On RAGTruth, the label-free synthetic support axis (ours) sits below NLI and below plain lexical overlap, the strongest baseline. Among the *internals*-based scores only the in-domain probe, which uses RAGTruth human labels, clears NLI. Axis starts at chance (0.5).

signal	Data2txt	QA	Summary	macro
synth- d_S	0.487	0.611	0.658	0.585
confidence	0.576	0.544	0.679	0.600
NLI (whole-resp.)	0.547	0.567	0.633	0.582
Sent-NLI (mean)	0.611	0.557	0.620	0.596
Sent-NLI (min)	0.521	0.591	0.626	0.579
overlap	0.605	0.686	0.649	0.647
in-domain [†]	0.766	0.680	0.673	0.706
faithful-rate	0.357	0.823	0.795	—

Table 6: **C4, per task type.** Within-task RAGTruth AUROC; synth- d_S , confidence and in-domain are means over the three models, NLI and overlap are response-intrinsic. The pooled numbers in Table 4 overstate every signal because faithful base rates differ sharply across tasks (bottom row): overlap 0.753 $\rightarrow$ 0.647, NLI 0.673 $\rightarrow$ 0.582, the axis 0.596 $\rightarrow$ 0.585. *Sent-NLI* is the SummaC-style sentence-level baseline. Within a task every label-free signal is weak (0.58–0.65 macro; the axis is below chance on Data2txt), and overlap is merely the least-bad. [†]uses RAGTruth human labels (0.706 macro).

Limitations

We are explicit about what we do *not* show. (i) Within the answer-present stratum the decodable contrast is question–answer compatibility (a no-context ablation leaves it almost unchanged, raw 0.95–0.98) and coincides with our synthetic factuality label by construction, so we cannot claim a context-grounding signal there; only a question-fixed cross-strata control (0.66–0.88) shows any context-sensitive component. (ii) Cell B has zero yield, so the fully-orthogonalized ($corr \approx 0$) design

is not instantiated and all analyses use cells A/C/D, where $\text{corr}(S, O)=0.50$ over all 3,593 rows and 0.565 on the collision-free subset that Table 1 and the mid-layer band use; the realized data is a three-cell partial factorial, not a balanced 2×2 , and within $O=1$ the support and synthetic factuality labels are the identical vector, so $\text{supp}_{O=1}$ is simultaneously a factuality AUROC. (iii) The cosine alignment is measured on shared stimuli, so it shows alignment of a shared low-rank contrast, not a fully unpaired transfer. (iv) We evaluate a deterministic strided 2,000-row subset of the 2,700-row wandb/RAGTruth-processed test split (450 source passages, 4–5 responses each), covering three task types and one annotation scheme, and the negative is on 3/4 families (Mistral’s RAGTruth extraction OOM’d at the longer context length and we did not retry it at a smaller batch size). (v) All probes read the residual stream at the *final answer token*. CtrlPairs answers average 1.7 tokens, so this is near-lossless there; RAGTruth responses average 126.6 tokens, so last-token pooling discards most of the response. We extracted no other pooling and cannot test the alternative here, so the RAGTruth results should be read as specific to this choice. All four instruction-tuned models are also probed with a single raw completion template rather than their chat templates; this sensitivity is untested. (vi) Support labels are validated automatically and by a 40-item manual audit of clean cell C (0/40 mislabeled), not by a multi-annotator study with agreement statistics. (vii) Stronger internals detectors (e.g. ReDeEP (Sun et al., 2025)) and sampling-based methods (Manakul et al., 2023) are out of our single-A100 budget, so we show that *this* synthetic axis fails, not that internals cannot work.

Ethical Considerations

This paper reports a negative result about a class of proposed detectors, and its main practical risk is over-trust in the opposite direction: a label-free internals probe of the kind we study reaches only 0.54–0.66 AUROC on RAGTruth, below plain lexical overlap (0.753), so deploying one as a hallucination guard would give a false sense of coverage. We therefore recommend validating an internals-based grounding probe against a lexical-overlap baseline on in-domain human labels before claiming it is deployable. CtrlPairs is constructed automatically from NQ-Swap Wikipedia text and involves no new human subjects, no new annotation, and no

personally identifying information beyond what the source corpora already contain; the “wrong-question” items are deliberately mislabeled stimuli for probing and are not intended as training data. All experiments are inference-only on a single A100-40G, with no model training or fine-tuning.

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A Implementation Details and Reproducibility

Code, the constructed datasets, every result JSON and the figures are available at <https://github.com/Beiciccc/GroundLM>, together with a manifest mapping each number, table and figure in this paper to the script that produces it and the files that script reads and writes. The 2.4 GB of cached residual-stream features the analysis consumes are too large to distribute with the source: `reproduce.sh extract` regenerates them from the released datasets on one A100-40G, after which `reproduce.sh` recomputes every reported number, table and figure on CPU. Below we fix every choice a reader would need to re-run the study.

Layer choice and fold-split sensitivity. Table 1 fixes the analysis layer a priori at relative depth 0.5. Averaging the same collision-free, source-grouped estimate over every cached layer in the 0.4–0.6 band gives $\text{supp}_{O=1}$ of 0.990 ± 0.004 (Qwen, 5 layers), 0.996 ± 0.002 (Mistral, 7), 0.996 ± 0.001 (Llama, 7) and 0.998 ± 0.001 (Gemma, 9), and confidence-purged values of 0.765 ± 0.016 , 0.877 ± 0.003 , 0.792 ± 0.003 and 0.809 ± 0.008 : the result is not a single-layer artifact. We also avoid `GroupKFold`, which assigns folds by sorting group sizes: our group sizes are almost all tied (1,193/1,200 CtrlPairs items have three rows; every RAGTruth response is its own record), so its split depends on `numpy.argsort` tie-breaking and changes with the library version: we measured the Llama in-domain probe moving from 0.815 to 0.824 across two `numpy` releases on otherwise identical inputs (recorded in the protocol note at the top of `scripts/_cr_common.py`). Every reported estimate instead uses an explicitly seeded group splitter over source groups. The pooled probe estimates in Tables 1 and 4 are means over ten such splits (split-to-split $\text{sd} \leq 0.003$ and ≤ 0.007 respectively); the per-task breakdown in Table 6, the cluster-bootstrap intervals and Figures 2 and 3 are computed on the first split.

Parametric-belief check. Before probing we verify closed-book that each model actually prefers the gold answer: with the context re-

moved, we compare the mean answer-token log-probability of a_{true} against a_{swap} per item. The gold answer wins for 84.0% (Qwen), 92.6% (Mistral), 94.8% (Llama) and 95.2% (Gemma) of the 1,200 items. All reported results use the full set; the per-item flag is released in the `runs/*_v2/parametric_belief.npz` caches and the analysis library accepts a `knows-gold` mask, but no script in the released pipeline conditions on it.

Artifacts and licenses. We build on the entity-substitution framework of Longpre et al. (2021), whose own release is under the Apple Sample Code License, via the `pminervini/NQ-Swap` distribution (MIT) that our build script loads; RAGTruth (Niu et al., 2024) and the DeBERTaV3 MNLI/FEVER/ANLI checkpoint (Laurer et al., 2024; He et al., 2023) are MIT. The probed checkpoints are Qwen2.5-7B-Instruct and Mistral-7B-Instruct-v0.3 (Apache-2.0), Llama-3.1-8B-Instruct (Llama 3.1 Community License; this work is “Built with Llama”) and Gemma-2-9B-it (Gemma Terms of Use). All use is non-commercial academic research, consistent with the intended use each artifact states. Our code is released under the MIT license; the constructed CtrlPairs data derives from that NQ-Swap distribution and inherits its terms. The redistributed RAGTruth subset contains third-party source text that we did not filter for incidental contact details already present upstream.

Prompt and answer span. Activations are extracted from a raw template (no chat formatting): `"{context}\n\nQuestion: {question}\nAnswer:",` to which the *asserted answer*, a teacher-forced gold surface string (NQ-Swap $a_{\text{true}}/a_{\text{swap}}$ for CtrlPairs; the model-generated response for RAGTruth), is appended with a single leading space and no special tokens. Inputs are *left*-padded, so the answer occupies the final positions. The probe feature `lastL` is the last-token residual state at layer L (under left padding, the final answer token). This is the *only* pooling we extracted: the released caches contain no answer-span mean, so the sensitivity of our results to pooling is untested (Limitations).

Confidence scalars. Computed over the answer span only (logits shifted by one to predict the answer tokens): *mean log-prob* (mean gold-token log-probability), *mean max-softmax* (mean per-position top-1 probability), and *first max-softmax* (top-1

probability of the first answer token). The confidence confound set in the purge comprises three scalars: mean log-prob, mean max-softmax, and first max-softmax. In the no-context and shuffled-context ablations (Table 2) these scalars are re-measured under the ablated prompt, so each +conf column is purged against the confidence the model actually expresses in that condition.

Lexical overlap. Two quantities, both case-folded. `OVERLAP` (continuous, `lexical_overlap`) is the fraction of whitespace-tokenized answer tokens that occur as a substring of the lowercased context; `OVERLAPmeasured` (binary, the cell label) is whether the *whole* stripped lowercased answer string is a substring of the lowercased context. A cell is dropped when its measured presence disagrees with its intended overlap label.

Construction and the missing cell B. Ctrl-Pairs cells: $A = (q, c_{\text{org}}, a_{\text{true}})$; $C = (q', c_{\text{org}}, a_{\text{true}})$ with q' a donor question from another item; $D = (q, c_{\text{sub}}, a_{\text{true}})$ with an entity-swapped context. Cell B = $(q, \text{paraphrase}(c_{\text{org}}), a_{\text{true}})$ would be supported but answer-absent; it is produced only when a paraphraser is supplied and an item passes a *dual* filter: token-overlap of the answer with the rewrite ≤ 0.34 and NLI entailment of the answer by the rewrite ≥ 0.5 . Greedy rewrites that remove the answer string tend to fall below the entailment threshold, while those that stay entailed tend to retain the answer tokens; the released build supplies no paraphraser, so cell B has zero yield and experiments use cells A/C/D.

RAGTruth labels. Each (context, response) row gets one binary response-level label: `FAITHFUL=1` iff the dataset’s `hallucination_labels_processed` dictionary sums to zero (no flagged span). We use no sentence-level mapping; the sentence-level SummaC baseline (§5) decomposes the response at *scoring* time only. The same 2,000 fixed responses are scored by every probe model (hidden states/confidence extracted by teacher-forcing each response through that model), which is why the NLI and lexical-overlap columns are identical across models. The whole-response NLI baseline forms its hypothesis by declarativizing each (question, answer) pair via a fixed template and scores the entailment probability with the context as premise; the sentence-level variant instead uses

each response sentence as a hypothesis (§5).

Probing and cross-validation. Directions are mass-mean (difference of class means) unless stated otherwise; the in-domain RAGTruth probe and the logistic arms of Table 5 are L_2 -logistic with $C=0.5$ on standardized features. Within-stratum decodability uses 5-fold grouped cross-validation with the groups defined by *source*, not by record. For CtrlPairs the group is the normalized (question, context, answer) triple of the underlying NQ-Swap QA: `ITEM_ID` indexes one swap record, and because several records share a source QA the 1,200 cell-A rows cover only 555 distinct triples, so grouping on `ITEM_ID` left 903 rows with an identical twin in another fold. For the in-domain RAGTruth probe the group is the normalized source passage: the 2,000 released responses come from 450 passages (4–5 each), while `ITEM_ID` is a per-response counter, so the earlier grouping was row-level. Confidence and surface purging is fitted *within* each training fold and applied to that fold’s test rows. Cross-family transfer fits an orthogonal Procrustes map on shared stimuli at each model’s a-priori layer (nearest cached layer to relative depth 0.5), with the calibration/evaluation split also grouped by source.

Evaluation rigor. Moving from `ITEM_ID` groups to source-QA groups changes the C1 estimates by at most 0.007 (raw by ≤ 0.0005), and moving the RAGTruth probe from response groups to passage groups changes it by at most 0.008; both leaks were real but numerically inert, because a difference of class means in $\sim 3,500$ dimensions is shifted only $O(1/n)$ by a duplicated row. We resolve axis polarity on the train (or calibration) fold only and score the test projection as-is: the test-fold $\max(\text{AUC}, 1-\text{AUC})$ convention floors a non-predictive direction above 0.5 (it inflates a true-chance random direction to $\approx 0.54\text{--}0.59$), so no reported number uses it. The C1 analysis layer is fixed a priori at relative depth 0.5 (as C3/C4 already do) rather than `argmax`-selected on the reported metric. Tables 1, 4 and 6 (bar its Sent-NLI rows) are produced by `scripts/cr_final_tables.py`; the remaining reported quantities come from the scripts listed in `MANIFEST.md`, which maps each table, figure and quoted value to its producer and inputs.

Compute. All activation extraction runs on a single A100-40G; per-model batch sizes are re-

duced for large-vocabulary models to avoid OOM (Qwen 4, Gemma 2, Llama 6, Mistral 16). Parameter counts are 7.6B (Qwen2.5-7B-Instruct), 7.2B (Mistral-7B-Instruct-v0.3), 8.0B (Llama-3.1-8B-Instruct) and 9.2B (Gemma-2-9B-it); no model is trained or fine-tuned, so the budget is inference only. It totals 58,716 forward passes: per family, 3,593 CtrlPairs statements, 2,400 closed-book parametric-belief scorings ($1,200 \text{ items} \times \text{two candidate answers}$) and 3,593 each for the no-context and shuffled-context ablations, plus 2,000 RAGTruth responses for three families. All extraction ran on one A100-40G within a single day; our extraction logs do not instrument per-run wall-clock, so we report the forward-pass budget rather than a GPU-hour figure. All analyses, the sentence-level NLI baseline, and figures run on CPU/Apple-MPS without a GPU. The Mistral RAGTruth extraction was the one run that did not complete: it reuses the batch size tuned for the shorter CtrlPairs prompts (16) at the longer RAGTruth context length (2048 tokens) and exhausts memory, which is why C4 covers 3/4 families. Re-running it at the batch size used for Qwen would fix this; we report the negative on the three families we extracted.